

SI-WP-011

The Seat Without the Bench: Expertise-Reproduction Failure and the Precondition the Non-Removability Prescription Presupposes

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Methodological positioning: This paper presents a structural argument grounded in the published Synthience corpus, in cited empirical and theoretical work from third-party research, and in the architectural implications that follow from both. It is a precondition-critique of SI-WP-009: it does not dispute that paper's diagnosis or its prescription, and it strengthens rather than weakens the case for non-removable substantive human involvement by identifying a mechanism through which that prescription can be satisfied on its own terms while the condition that gives it meaning over time is quietly voided. The argument is conditional throughout. Its antecedent, that AI is severing the entry-level pathway through which professions form their future experts, is inherited from SI-WP-009 and SI-WP-010, is empirically contested, and is not relitigated here; the paper specifies what follows if the antecedent holds and does not assert that severance is presently established. Its central forward claim, that practitioners formed under conditions that remove the formative apprenticeship rep will fail to acquire the production-class competence the non-removable seat requires, is a named and increasingly discussed risk construct in the professional-education literature, with sparse, early, and mixed empirical support; it is predicted from the structure of expertise formation, not observed, and the paper says so at full strength wherever the claim appears. The structural mechanism and the conditional in which it is stated are asserted at strength; the antecedent and the forward outcome are calibrated to their evidence. The failure the paper describes is not invisible in principle: it is invisible to the instruments that measure present performance rather than formation trajectory, and visible to an instrument of a kind most fields do not yet run. The paper specifies the conditions under which it is wrong.

Abstract

SI-WP-009 (Gantz, 2026) argues that competitive optimization in advanced AI development produces a default trajectory toward the removal of substantive human contribution from consequence-present systems, and prescribes architecture that makes substantive human involvement structurally non-removable. SI-WP-010 (Gantz, 2026) instantiates one binding instrument, licensing-as-architecture, operating at the function-definition layer. Both papers answer the question of how to keep a substantively capable human in the consequential loop. This paper asks a prior question that the prescription presupposes: who will be capable of filling the seat the prescription preserves. Expertise is modeled, following standard human-capital and internal-labor-market economics, as a stock (the currently

competent practitioners) replenished by a flow (the apprenticeship pathway that converts novices into capable seniors). The contribution is not that framing, which is inherited; it is the identification of a coupled failure structure within it, whose distinctive feature is a triple identity: the resource that masks the severed flow is numerically the same resource being depleted, and is also the resource a field would need to rebuild the flow. The paper argues that if entry-level automation severs the flow while the stock remains intact, the severance is masked by present-tense operational metrics (output, oversight pass-rates, present staffing) because the practitioners who supply those metrics are the stock the flow exists to replenish; and that the most expensive-to-reconstruct component of the teaching capacity needed to rebuild the flow, the calibrated intervention judgment embodied in active practitioners, depletes on the same clock as the mask, so that the cost of restoration rises as the window for it closes. The paper states this as a restoration-window inequality, not as irreversibility: the flow can be rebuilt, but plausibly only at a cost in time and institutional capacity exceeding what the remaining stock can supply once the deficit becomes visible. The relevant unit is the formation ecology, the firms, training institutions, supervisors, and practice settings that together supply a class of production-capable practitioners, and the mechanism binds only where severance is ecology-wide; where an unaffected external reservoir remains, the failure is a hiring problem, not expertise-reproduction failure. The argument extends the observability-failure lane the corpus has established (SI-WP-007, invisible maintenance; SI-WP-008, ceremonial governance; SI-WP-009, distributed harm), all of which describe present-tense invisibility, by identifying a temporally displaced, self-masking instance: the masked variable is a future non-event, and the mask is the very stock the flow would replenish. It remains distinct from SI-WP-010 by addressing non-reproduction of the future flow rather than extraction from the present stock. The paper's hinge is whether the production-class competence the seat requires can form through verification alone, without first-hand production experience; it argues, as a domain-bound and falsifiable prediction grounded in transfer-appropriate processing, that where the human fallback must intervene under conditions the AI may be absent for, verification-only formation does not build that competence, while conceding, with the worked-example literature, that in well-structured domains verification-like study builds competence effectively. The paper's sharpest claim is that the corpus's own binding architecture can audit clean while failing: a licensing instrument that measures present substantive engagement passes for exactly as long as the intact stock can answer it. This is not a defect in that instrument relative to its stated object; it is an architectural insufficiency: the instrument verifies the seat and is silent on the bench. The failure is not undetectable in principle: it is invisible to the instruments most fields actually deploy and visible to a formation-trajectory instrument most do not yet run. The non-removability prescription is necessary and, on its own, temporally insufficient; the paper specifies the design requirement for the missing instrument, one that measures the formation-trajectory of the bench rather than the performance of the stock, and defers its full architecture. The displacement antecedent is inherited and contested; the non-formation claim is a named, sparsely evidenced risk construct; the structural mechanism is asserted at strength; and the paper specifies the conditions under which it is wrong.

Core claim (conditional): *If AI automation severs the entry-level pathway through which a profession forms its future experts, then the architecture that makes substantive human involvement non-removable preserves the seat but not the pathway that produces humans capable of filling it. The severance would be masked by present-tense operational metrics for as long as the existing expert stock remains capable of supplying them, not because the failure is undetectable in principle, but because the instruments most fields deploy measure present performance rather than formation trajectory, and restoration would grow more costly as the same stock needed to detect, teach, and rebuild the pathway turns over. On this conditional, the non-removability prescription is necessary and temporally insufficient: a sufficient architecture must add an instrument that measures the formation of future expertise rather than the performance of present expertise, because present performance is the variable the severance*

does not touch until the stock that supplies it has thinned. The paper does not assert that the antecedent currently holds.

Keywords: *expertise reproduction, stock and flow, formation ecology, apprenticeship, situated learning, internal labor markets, ironies of automation, observability failure, non-removability, human-in-the-loop, verification competence, transfer-appropriate processing, schema acquisition, AI governance, pre-empirical analysis*

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1. The Precondition the Prescription Presupposes

SI-WP-009 names a trajectory and prescribes an architecture against it. The trajectory is the progressive removal of substantive human contribution from the consequential loops of AI-mediated systems, driven not by intent but by competitive optimization acting on human judgment, oversight, and authority as removable costs. The prescription is architecture that makes substantive human involvement structurally non-removable, and that makes compliance the path of least resistance under the same competitive pressure that otherwise drives removal. SI-WP-010 develops one instrument that satisfies the prescription at a specific layer: licensing-as-architecture, which binds what an institution is authorized to claim about its work product to the substantive human engagement constitutive of that claim, so that an institution authorized to produce regulated professional work must have a qualified professional substantively engaged in producing it, not signing off on outputs from which they have been progressively excluded.

Both papers answer the same question at different scopes: how to keep a substantively capable human in the consequential loop. This paper asks a question that sits prior to that one and that the prescription presupposes without stating. The prescription preserves a seat: a place in the loop that a substantively capable human is required to occupy. It presupposes that substantively capable humans exist to occupy it. That presupposition holds today, because the current stock of experts was formed under conditions that may no longer obtain for the people who would replace them. The question this paper asks is not whether the seat can be preserved. It is who will be capable of filling it.

The thesis is not that AI takes junior jobs, and it is not a forecast of unemployment. It is a conditional claim about reproduction. *If* the pathway producing future occupants of the non-removable seat is severed underneath the prescription, then the prescription preserves a seat for which the supply of qualified occupants quietly runs out, and the severance would be invisible to the instruments most fields deploy to verify compliance, though not, as §3 and §8.3 make precise, invisible in principle to an instrument that measured formation trajectory instead of present performance. The paper develops this conditional and specifies what would have to be measured to detect its antecedent before the consequence arrives. It does not assert that the antecedent currently holds; that question belongs to SI-WP-009 and SI-WP-010 and to the contested empirical literature those papers engage, and it is taken up only far enough, in §8.5 and §9, to state honestly what is and is not known.

This is not a refutation of SI-WP-009 or SI-WP-010. It is a strengthening of the first by identification of a way the prescription can be satisfied while failing. On the conditional, a field could hold a substantively capable human in every consequential seat, pass every present-engagement audit, post improving productivity, and be in expertise-reproduction failure at the same time, the failure invisible to the instruments that read present performance, though not, as §3.1 and §8.3 make precise, invisible in principle, with the first signal *to operational metrics*

arriving only when the field reaches for the next generation of experts and finds the bench unfilled, and arriving earlier, to a field that measured formation trajectory, well before then. SI-WP-009 establishes that the seat must not be removed. This paper establishes that preserving the seat is not the same as preserving the supply, and that the corpus has so far specified instruments only for the former.

The fields where this matters most are not the heavily credentialed professions but the large body of professional knowledge work (software engineering, data and quantitative analysis, consulting, and the analytic functions of corporate professional life) where there is no licensure gate, no mandatory unaided examination at entry, and no residency, and where the pathway from novice to senior runs through exactly the lower-rung production work most exposed to automation. These are the fields this paper centers; the heavily credentialed professions enter later (§8.4, §8.6) as the bounding contrast that shows what preserving the flow costs and which instruments can catch its loss. The paper is a rapid-response standalone in Vertical 2, a sibling to SI-WP-009 (the field-scale diagnosis it is conditioned on) and SI-WP-010 (the labor-layer manifestation it sits beside, whose licensing boundary binds the regulated professions and thereby leaves professional knowledge work unbound, the layer this paper occupies). It inherits their displacement premise as a contested antecedent rather than re-establishing it, and it is written for the same readers with leverage at the architectural decision points: policymakers and governance architects whose binding instruments could, if they are designed only to measure present engagement, certify a field as compliant for exactly as long as its existing stock of formed experts lasts.

2. Stock and Flow: A Borrowed Frame and the Coupling Inside It

Expertise in a field is two things, not one, and the two are separately observable. The first is a **stock**: the set of currently competent practitioners, the seniors who can produce the work, catch the errors, and intervene when a system fails. The second is a **flow**: the pathway that converts novices into members of that stock over time, through the apprenticeship, the graduated exposure to consequential work, and the situated learning by which a person becomes someone the field can rely on. The stock is what a field has now. The flow is how a field replaces what it has as the stock retires, leaves, or dies.

The vocabulary and the underlying economics are borrowed, and the paper claims no originality for them. That jobs are simultaneously sites of production and sites of learning, that the labor market is, in Rosen's (1972a) formulation, an implicit market for learning opportunities dual to the market for jobs, with learning a joint product of work (Rosen, 1972b), is established human-capital theory. That entry positions function as the ports through which workers enter an internal labor market and ascend its promotion ladders, acquiring through workplace exposure the capability the ladder presupposes, is the core of Doeringer and Piore (1971). That production itself generates capability, that doing the work is how the capacity to do the work is built, is the learning-by-doing tradition (Arrow, 1962). None of this is new, and the paper does not present stock-and-flow modeling as a contribution. It is the expository frame, and it is retained because it makes the contribution legible in a single picture.

The contribution is a coupling that this established frame contains but has not been used to name, because the conditions that activate it did not previously obtain. The internal-labor-market literature assumes the developmental pathway persists; its concern is whether an organization uses the pathway well, screens through it accurately, allocates along it efficiently. The coupling this paper identifies is what happens when the pathway is not used badly but *removed*, when the lower-rung work that was simultaneously output and developmental substrate is automated as human work, and three properties hold together: the flow is severed;

the present stock masks the severance because the stock supplies the metrics a field's ordinary instruments consult; and the same stock is the irreplaceable component of the teaching capacity a field would need to rebuild the flow once the deficit is felt. The claim is not "stock and flow exist." It is that these three properties, coupled, produce a failure that is self-masking and whose cost of correction rises as the window for correction closes: a structure that does not appear when the pathway is merely used poorly, and that the existing literature therefore has had no occasion to model.

This coupling marks the seam between this paper and SI-WP-010. SI-WP-010 documents extraction *from the present stock*: the function an existing worker performs is redefined under competitive pressure into something the worker is no longer needed for, and the worker is displaced or devalued. Its cases (the engineers instructed to record their workflows before being laid off, the data workers training their own replacements, the supervisor offered a steep pay cut and then terminated) are all cases of value extracted from people who are already in the stock. SI-WP-010's nearest approach to this paper's territory is its observation that senior judgment "becomes visible only after it has been removed": that is a masking observation, but about the present stock: the value of the seniors who were there. This paper takes up the orthogonal question SI-WP-010 does not: not what happens to the seniors who are there, but where the next seniors come from. SI-WP-010 also confined its licensing instrument to regulated professions; the layer that instrument leaves unbound, professional knowledge work without a credentialing gate, is this paper's center of gravity. One paper is about taking; this one is about not-replacing. One is about the present stock; this one is about the future flow.

The coupling also marks what SI-WP-009's prescription does and does not bind. Non-removability binds the stock: it requires that a capable human occupy the loop now. It is silent on the flow. A prescription that binds the stock while the flow is severed does exactly what it says, keeping a capable human in the seat, and accomplishes nothing for the supply of capable humans, because supply is a property of the flow, and the flow is not what the prescription measures.

3. The Mechanism: A Self-Consuming Mask and a Restoration-Window Inequality

This is the paper's structural core. The argument is stated as a conditional and its force does not depend on the antecedent being presently true; it depends only on the structural relations holding *if* the antecedent obtains. The argument has three parts: the severance would be masked; the mask would be composed of the resource being depleted; and restoration would grow costlier than the window allows, not because the flow cannot in principle be rebuilt, but because the conditions for rebuilding it degrade on the same clock that reveals the need.

3.1 The masking dynamic

Suppose the flow is severed: suppose the formative work through which novices became seniors is now performed by AI rather than by novices. The masking requires one further condition, which should be named rather than assumed: that AI performs the displaced formative work *well enough that aggregate output does not visibly degrade*. This is an empirical premise about tool quality, not a structural entailment of severance, and stating it is a strengthening, because its failure is favorable: where AI does the formative work visibly worse, operational metrics drop, the severance is self-revealing, and the field simply stops automating that work. The masking dynamic is therefore scoped to the case where the tool is good enough to keep the metrics green, which is also the case worth worrying about.

Under that condition, the stock does not change at the moment of severance. The current seniors, formed under the prior pathway, continue to perform; with AI assistance they may perform better. Output quality holds or improves. Oversight functions pass, because the seniors are competent verifiers of the AI's work. Productivity rises, because the seniors plus AI out-produce the seniors alone, and because the cost of the novices who used to occupy the formative rungs has been removed. The present-tense operational indicators most commonly deployed (output, quality pass-rates, throughput, current delivery, senior-review success) are green or improving, because they are supplied by the stock and do not measure the formative trajectory of the bench.

One staffing variable needs separating from the rest, because the paper's own antecedent implies it can move. *Total* headcount, junior-role counts, and labor-cost composition can remain stable or even improve: bodies remain in junior-labeled roles, contractors substitute, roles are relabeled around AI-mediated work, senior productivity rises. What changes is *formation-bearing* headcount: the number of entrants receiving work that carries the formative features of §5.1. The relevant variable is not headcount as such but formation-bearing headcount, and total staffing surfaces can read stable while the formation-bearing portion of the pathway collapses underneath them. A field watching role-occupancy and current output sees neither the collapse nor its leading edge.

The boundary here must be drawn precisely, because the broad version of the masking claim is false. The severance is invisible *to operational metrics*, the measures of present output and present staffing, because those metrics are supplied by the stock, and the stock is the variable the severance does not yet touch. It is not invisible to every organizational instrument. Succession planning, talent review, and the family of forward-looking assessments that plot present performance against future potential are designed precisely to look past current output, and a field that runs them seriously can in principle see a thinning bench. But the masking against *those* instruments is not absence of measurement; it is an epistemic decoupling, addressed in §8.3, in which the proxy a forward-looking instrument reads, a junior who excels at verifying AI output, stays highly correlated with past indicators of potential while becoming decoupled from the target trait, production-class competence under failure, that it is supposed to predict. The defensible claim is therefore narrow: operational performance metrics cannot detect the severance, and forward-looking instruments detect it *only if* they measure demonstrated developmental trajectory under conditions where the AI is absent or failed: whether the present, productive juniors are becoming people who can do what seniors do when the tool is gone or wrong. Many such instruments do not measure that; they count bodies in junior roles and read those bodies' current output, both of which look healthy. The failure is not invisible in principle. It is invisible to the instruments that measure present performance, and visible to one that measures formation trajectory under unaided-or-failed conditions, an instrument most fields do not run. The claim is about which instruments are deployed, not about an inherent invisibility no instrument could pierce.

3.2 The mask is the depleting stock

This is the structurally distinctive feature, and it is what separates this failure mode from ordinary pipeline mismanagement. The thing that hides the severed flow, the competent seniors whose performance keeps the operational metrics green, *is the stock the flow exists to replenish*. The mask is made of the inventory being drawn down. It is not an external screen placed over the problem; it is the problem's own remaining substance, performing.

That self-consuming structure determines the failure's timing, and the timing depends on a production-function shape worth naming rather than assuming. If output degrades gracefully

and linearly as the stock thins, the operational signal arrives relatively early and the stock is not yet largely gone. If output holds until the stock nears exhaustion and then drops, a cliff-shaped function, plausible where the remaining seniors plus AI can carry the load until they cannot, the operational signal arrives late, when the stock that composed the mask is largely gone. Which obtains is one of the quantitative terms of §3.3, not a structural given. In the cliff-shaped case, and to the degree the function approaches it, the signal that the pathway needed rebuilding arrives at the same time as the loss of the resource that could most cheaply have rebuilt it. The point is not that the timing is uniformly catastrophic; it is that the worse the masking (the more cliff-shaped the function), the more the detection signal and the loss of the means to answer it converge on the same moment, and operational metrics give no advance warning of which regime a field is in.

3.3 The restoration-window inequality

A masked failure that arrived late but remained cheaply correctable would be a management problem, not a structural one. The claim is not that the flow is impossible to restore. A field can rebuild training ladders, stand up simulation, create residencies, deliberately preserve formative work as human work, pay seniors to teach, or, where an unaffected external reservoir exists, import expertise. Restoration is possible. The claim is an inequality: *if* the flow is severed and the severance is masked until stock-turnover, the cost of restoration, in time and institutional capacity, plausibly exceeds what the remaining stock can supply within the window before it is gone. Three terms drive the inequality.

First, the formative periphery has been removed as human work, not merely reassigned, so rebuilding the flow requires re-creating that work, and re-creating it means re-incurring the cost the automation was adopted to remove. This is the difference between a *thinned* pipeline, which can be widened when the need is felt, and a *severed* one, where the rung a widened pipeline would carry no longer exists as human work and must be reconstructed against the economics that eliminated it.

Second, the institutional knowledge of *how* to form a senior, the supervision practices, the graduated-exposure structures, the tacit craft of bringing someone along, atrophies when it is not practiced, so a field that decides to restart the flow after a generation may find it has forgotten the method.

Third, and decisively, the teaching resource is not exhausted by the active stock, but its irreplaceable component depletes with the stock. Teaching capacity is broader than the active senior practitioners: it includes curricula, training institutions, standards, simulations, recorded cases, credentialing regimes, and retired practitioners. The claim is not that all of this disappears when the stock thins. It is that one component of it, the calibrated intervention judgment embodied in active or recently active practitioners, the living capacity to demonstrate, supervise, and correct production-class intervention *under contemporary conditions*, is the most expensive to reconstruct and is the component that depletes on the same clock as the mask. The persistent institutional apparatus is not an equivalent substitute when the work has changed: when AI alters what the formative reps must be, a field needs active practitioners who understand the new frontier to redesign the curriculum, the residency, the simulation; an apparatus cut off from current production-grade judgment is left teaching the prior generation's pathway. And in the fields this paper centers (software, data, analysis, consulting, professional knowledge work) there is no large institutional apparatus at all: no residency, no board, no simulation program. There, the teaching resource simply *is* the senior at the next desk, and the self-consuming mask is literal: the people whose departure reveals the deficit are the only people who could have taught the replacement pathway.

These three terms make restoration cost rise as the window to pay it closes. The inequality is not a claim of impossibility; it is a claim that, on the conditional, the field passes the point at which restoration is affordable before it can see that the point existed. Whether the inequality binds in any given field is an empirical and quantitative question, depending on turnover rates, on the production-function shape of §3.2, on how much of the periphery was eliminated versus relocated, on whether teaching capacity was preserved, and on the scope condition of the next paragraph, and the paper does not claim to settle it in general. It claims that the inequality is the right object to measure, and that operational metrics measure none of its terms.

The inequality binds at the scale at which the severance is ecology-wide, and that scope must be pinned because bindingness depends on it. The relevant unit is not any single employer but the *formation ecology*: the firms, training institutions, supervisors, credentialing bodies, and practice settings through which entrants of a class acquire production-class competence. If an individual firm severs its pipeline while an intact external ecology persists, the firm has a hiring problem, not expertise-reproduction failure: it draws from the reservoir, and the third term does not bind. The inequality is a claim about severance at the scale of the ecology, where there is no unaffected reservoir to draw on. International hiring, adjacent-field transfer, retired practitioners, and preserved training institutions are real escape hatches; they defeat the mechanism wherever they are large enough, timely enough, and formation-compatible enough to reproduce the specific intervention competence at issue. The mechanism's claim is precisely that in some ecologies, those whose competence is field-, jurisdiction-, or safety-specific, or tacit enough that import and cross-field transfer cannot quickly replenish it, the external reservoir is itself downstream of the same severance, and no unaffected stock remains. The reservoir defeats the mechanism unless the reservoir is failing too; that is the boundary the inequality turns on at ecology scale, and it is the same demand-side condition §8.1 frames at population scale: a smaller needed stock and an unaffected reservoir are the two ways the severance fails to bind.

That the inequality *can* be beaten by deliberate, expensive, sustained institutional effort is not a counterexample to it; it is the inequality's confirmation, and §8.4 develops the clearest case: the professions that have in fact paid the cost.

3.4 Bainbridge, Scaled, and What Is New at the New Scale

Bainbridge's *ironies of automation* (1983) established two results at the level of the individual operator: automating the routine parts of a task degrades the operator's skill at the non-routine residual, and the operator is needed precisely when the automation fails, which is the moment their degraded skill is least adequate. Scaling that structure from the individual to the field is not, by itself, a contribution, and the paper does not claim it as one. "More operators deskilling" is Bainbridge with a larger sample.

What is new at the field scale is a property absent at the individual scale: the loss of the irreplaceable teaching component changes restoration from a retraining problem into the inequality of §3.3. At the individual level, a deskilled operator can be retrained, because the field still contains other operators who retain the skill: the teaching resource exists outside the individual. At the ecology level, if the flow remains severed until the stock turns over, the calibrated-judgment component of the teaching resource does not exist in an unaffected reservoir outside the ecology, because it was the ecology's stock. The individual-level irony is recoverable by drawing on the field; the ecology-level irony, past stock-turnover and absent an unaffected reservoir, has no larger pool to draw on. That emergent property, together with the self-consuming mask of §3.2, is the contribution. Stock-and-flow and the scaled Bainbridge are the scaffolding that makes it visible.

4. The Observability Family, Extended Temporally

The corpus has established an observability-failure lane, and this paper extends it. Extension requires showing a feature the prior papers do not have, not restating their structure with a new noun; the difference is precise, and the prior papers' arguments are not reused here.

SI-WP-007 establishes invisible maintenance: the benefit of continuity work is diffuse and produces no signal when the work is performed correctly, so bounded-rational actors under-resource it. The invisible thing exists now and is invisible now. SI-WP-008 establishes ceremonial governance: formal oversight generates the appearance of oversight while being hollow, and the formal process destroys the reporting infrastructure that would reveal the hollowing. The oversight is hollow now; the signal is destroyed now. SI-WP-009 establishes the confidence trap under distributed harm: AI-mediated harm is statistically distributed, lagged, and confounded, so it never concentrates into a discrete correction-forcing event. The harm is occurring now and never crosses the salience threshold.

All three are forms of *present-tense* invisibility: a process happening now that cannot be seen now. The invisibility this paper describes is structurally different in two respects. First, it is **temporally displaced**: the invisible thing is not a present hidden process but a *future non-event*: a senior who will not be formed, a competence that will be absent from a person who does not yet occupy the seat. At the moment of severance there is nothing yet failing to point at; the failure is a non-formation that becomes a deficit only when the stock that masks it is gone. Second, it is **self-masking through a triple identity**, and this is the genuinely novel structure, sharper than the prior cases rather than merely different in flavor. It is not that prior invisibilities were externally imposed while this one is intrinsic: SI-WP-007's signal-lessness is intrinsic to the nature of maintenance, and SI-WP-008's self-destroying reporting infrastructure is intrinsic to the hollowing, so "external versus intrinsic" would not cleanly separate this paper from its predecessors. What separates it is that the masking variable is numerically identical to two other things at once: the resource being depleted, and the resource required for restoration. The mask is the inventory, and the inventory is the teacher. No prior case in the family has this triple identity, and it is the reason the failure both hides itself and consumes the means of its own repair on a single clock. This is the temporally displaced, self-masking instance of the observability-failure lane: a new invisibility structure in the same family, which is why it belongs in the corpus.

One boundary must be drawn sharply, because it is the difference between this paper and SI-WP-008's first mechanism. SI-WP-008 owns expertise *degradation*: the cognitive decay of skill in present practitioners through disuse, experts losing skill they had. This paper is about the opposite direction: novices never acquiring skill in the first place because the formative work was performed by AI during the period in which they would have acquired it. Degradation and non-formation are different phenomena with different evidence bases. This paper neither relitigates degradation nor rests on the deskilling literature that documents it; where that literature is invoked elsewhere in the corpus it supports SI-WP-008's mechanism, not this one. This paper's load is carried by the structure of formation, and its forward claim is calibrated in §9 to what the formation literature currently supports, which is that non-formation under AI is a named and contested risk, not a demonstrated effect.

5. The Hinge: Can the Seat's Competence Form Through Verification Alone?

There is a rebuttal that, if it holds, dissolves the paper, and the paper turns on it. The rebuttal is that the flow is not severed but *rerouted*: as entry-level production work is automated, a new

class of entry-level work appears, namely verifying, checking, and correcting AI output, and novices who do that work form into capable seniors just as the old apprenticeship formed them. If junior AI-verification roles replenish the flow, there is no severance and no crisis; there is a changed pathway with the same output. The argument reduces to one question: can the production-class competence the non-removable seat requires form through verification alone, without first-hand production experience?

5.1 What the formative rep provided

The apprenticeship pathway provided a specific kind of rep, characterizable precisely enough to test. The formative rep has four features. **First**, the learner generates a response or artifact *before* seeing the answer: the cognition runs forward, from problem to attempt. **Second**, the feedback is *consequential*: the learner owns the downstream result of the attempt, not merely an authority's correction of it. **Third**, the signal is *valid, timely, and unambiguous*: the conditions Kahneman and Klein (2009) identify as necessary for the formation of genuine expert judgment rather than confident judgment that was never calibrated. **Fourth**, there is *genuine productive struggle with self-generated error*: the learner makes their own mistakes, recognizes them as theirs, and works through them, the mechanism by which tacit knowledge (Polanyi, 1966; Eraut, 2000) is laid down through situated participation (Lave and Wenger, 1991).

5.2 The boundary the worked-example literature draws

The hinge cannot be "verification teaches nothing," because that is false, and the strongest evidence against it is worth stating against the paper's own thesis. Sweller's (1988) work establishes that for novices, unguided problem-solving can be an *inferior* learning device: means-ends problem-solving consumes cognitive capacity that schema acquisition needs, and studying worked examples, a verification-like activity of inspecting and comprehending a provided solution, can build schemas more effectively than struggling through production. In well-structured domains with clean solution paths and unambiguous standards, verification-like study works, and may work better than the struggle the formative rep romanticizes.

That literature draws this paper's boundary rather than defeating its hinge. The distinction it forces is not production-versus-verification but *schema-building support versus schema-bypassing automation*. A worked example builds a schema because it directs the learner's cognitive work toward the structure of the solution; AI that supplies a conclusion without process, to a learner who lacks the prior schemas to evaluate it, may still produce familiarity or pattern exposure, but does not reliably build the generative schema the intervention seat requires. The risk to formation is highest where AI supplies conclusions without process transparency, where the learner has insufficient prior schemas to evaluate the output, where the pathway does not require independent reasoning before AI exposure, and where the target competence is performance under uncertain or distribution-shifted conditions the AI may not cover. Where AI functions as a good worked example (transparent, explanatory, engaged after the learner forms a hypothesis, with valid feedback) Sweller is on the optimist's side and the paper's thesis does not fire. The hinge applies to the schema-bypassing case, not to AI assistance in general, and that boundary is the paper's, not a concession away from it.

5.3 Which competence does the seat require?

The seat does not require the same competence in every domain, and the hinge must be domain-bound or it overclaims. Three role types can be distinguished by what the human fallback must do when the AI's output is wrong. In **detection-and-escalation** roles, the human must recognize that something is off and route it elsewhere; the fix comes from another part of the system. In **diagnosis-and-escalation** roles, the human must localize the fault before

routing it. In **production-class intervention** roles, the human must supply what the failed system did not: take over, produce the work, repair or stabilize the system under uncertainty.

These are not bright lines, and a role can migrate between them as system design changes: an architecture that learns to fail gracefully, isolating errors and routing them to a central tier, can move a role from production-class toward diagnosis-and-escalation. So the typology needs an operational sorting criterion, not merely three labels, or the thesis can be dodged by reclassifying any inconvenient case. The criterion is this: a role is **production-class intervention** if, at the point of AI failure, the institutionally required human action is to generate or repair the work product under uncertainty with *no further escalation path that itself produces the fix*: the buck stops at this human for production, not merely for noticing. Detection-and-escalation is where the human need only flag for others to resolve; diagnosis-and-escalation adds localizing the fault. SI-WP-009's prescription is not satisfied by a human who can wave through correct outputs; it is satisfied by a human who can intervene when the system fails. Where that intervention is production-class by the criterion above, the seat requires production-class competence, and the thesis applies most strongly; where the role is genuinely detection-and-escalation, verification-style formation may suffice and the thesis does not strongly apply; the middle case is partial. This is a domain criterion, not a universal law, and the §5.4 transfer study draws its population from roles meeting the production-class test, so the prediction cannot be evaded by reclassification.

5.4 The transfer bet

A verification role can be designed to satisfy the *form* of the first three formative features: a junior can be made to commit a correctness judgment before ground truth is revealed (forward generation, of a judgment), to own the consequence of a bad approval (consequential feedback, where the safety net is removed), and to receive a valid, timely signal on those judgments. So the hinge cannot be that verification fails the first three features by construction; that would be unfalsifiable. The fourth feature, productive struggle with self-generated error, is where the difference concentrates, and the reason is that its *form* is mimickable while its *substance* differs: a verifier can struggle with their own judgment-errors, but that is not the same struggle as producing under uncertainty and owning the production-error. It is worth distinguishing three operations to locate the gap precisely: *recognition* (detecting an output may be wrong), *evaluation* (explaining why it is wrong or whether it meets a standard), and *generation/intervention* (producing a substitute or stabilizing a failure under uncertainty, with the artifact source absent or wrong). Verification builds recognition and evaluation. The hinge concerns transfer from evaluation to generation/intervention: not whether verification can teach, but whether what it teaches is the competence the production-class seat requires.

The prediction is grounded, not merely asserted. By the principle of transfer-appropriate processing (Morris, Bransford, and Franks, 1977), performance transfers between training and use to the degree the cognitive operations match; where the trained operation (evaluating a presented artifact) and the required operation (generating an intervention without a presented artifact) differ in kind, transfer cannot be presumed and must be demonstrated. The generation effect points the same way: material a learner produces is retained and structured differently from material a learner reviews (Bertsch et al., 2007). The boundary of §5.2 then localizes the prediction: where AI supplies conclusions without process, the schema-bypassing case, the learner runs no forward generation of the production artifact or intervention and builds no production schema, so verification experience does not transfer to production competence; where AI functions as a transparent worked example, §5.2 already concedes it can. The hinge is therefore the bounded claim that *in production-class-intervention domains, verification-only formation through schema-bypassing tools does not build the generation competence the seat*

requires, and the burden falls on the optimist to show that repeated verification builds the generative schemas production demands, the null, given transfer specificity, being that it does not. A cohort formed exclusively on such verification will, when the AI fails in a way that demands production, be unable to supply at the level the departed stock could, even if it verifies as well or better.

This is a bet with stated stakes and an explicit refutation condition: a transfer study showing that verification-trained practitioners intervene as competently as production-trained practitioners at genuine system-failure points, in production-class domains, refutes the hinge and with it the paper. Stated this way the hinge is neither rigged nor empty; it is a claim that could be false and that names what would show it false. Its current evidential status, named and discussed in the professional-education literature, sparsely and contestedly evidenced, is stated in §9 and not inflated here.

5.5 The consequence for the prescription

The hinge resolves a confusion that would otherwise protect the prescription. Even a verification role that is consequential and forward-generating, a *good* verification role, not a ceremonial one, preserves the **seat** while, on the transfer prediction, not preserving the **formative function** in production-class domains. It puts a present, engaged human in the loop and forms that human's capacity to recognize and evaluate outputs. What it would not do, on the prediction, is form the capacity to generate when the output source fails, which is the capacity the production-class seat exists for. The seat is occupied; the supply of people who can do what the seat requires is not being produced. That gap is invisible to anyone watching the seat, and visible only to an instrument that measured whether the occupants can produce under failure.

6. Why the Corpus's Own Prescription Can Audit Clean While Failing

This is the precondition-critique landing, and it is the paper's sharpest claim, in its strongest form where no robust credentialing gate sits between the entrant and the stock, which is this paper's center, and bounded for the gated professions by the rejoinder developed below. Two propositions must be kept distinct, because the section turns on the difference. The first is that the *seat-binding instrument reads clean*: a present-engagement check returns a truthful yes for exactly as long as the intact stock can answer it. That sub-claim is universal: it holds even in medicine, where the present-engagement check is satisfied by every genuinely-engaged senior. The second is that the *profession's full apparatus reads clean*: nothing in the field's instruments reveals the severance. That aggregate claim is what the rejoinder bounds, because a profession's credentialing layer is a non-seat-binding instrument that can catch the severance where the seat-binding instrument cannot. The section establishes the first proposition in general and then scopes the second.

SI-WP-010 specifies licensing-as-architecture: an institution retains its authorization to claim that it produces regulated professional work only if the substantive human engagement constitutive of that work is present. The instrument verifies the seat. It asks, of the work product: is a qualified professional substantively engaged in producing this? And on the conditional this paper develops, the intact stock answers yes: truthfully, every time, for as long as the stock lasts.

The masking dynamic of §3 therefore has a consequence the corpus has not stated about its own prescription: a licensing instrument that measures present substantive engagement passes while the pipeline is severed. It measures present substantive engagement; present substantive engagement is supplied by the stock; the stock is the variable the severance does not touch

until it turns over. A field could hold full licensing compliance, with every audit green and every required human present and genuinely engaged, for exactly as long as its existing stock of formed experts lasts, and the instrument would not reveal the severance, because the instrument measures the thing the severance leaves intact.

This is not a defect in the licensing instrument relative to its stated object. The instrument was built to verify present substantive engagement, and it does that correctly; calling it "failing" for not measuring future formation would be judging it against a surface outside its design. The point is an *architectural insufficiency*: the instrument verifies the seat and is silent on the bench, and a governance architecture that contains only seat-binding instruments is, as a whole, temporally insufficient even when each instrument works. This is the identification of a blind spot in the architecture, not a malfunction in SI-WP-010's component. The non-removability prescription is necessary: a field that removes the seat fails immediately and visibly, and SI-WP-009 is right that it must not be removed. But the prescription is, on its own, *temporally insufficient*: it preserves the seat for as long as the stock can fill it, and on the conditional it runs out of qualified occupants on a lag the prescription has no instrument to see.

There is an obvious rejoinder that sharpens the claim rather than defeating it, and it must be confronted because the word "licensed" already presupposes its subject. A licensing instrument requires *credentialed* professionals; credentialing is itself a formation instrument: in the heavily regulated professions, an entry examination and supervised progression that a candidate must pass before joining the stock. If the flow were severed and the credentialing examination tested production-class intervention under failure, candidate readiness would fall and pass rates would drop, and the credentialing system would register the severance well before the stock turned over. So for those professions the *aggregate* clean-audit claim is *false as a general claim*: the seat-binding instrument still reads clean, but the profession's credentialing layer does not, since that layer is a partial, real-world version of the §7 flow-instrument, and it would catch the severance early. This is the credentialing sensor, and it is why medicine notices in time (developed in §8.6). The aggregate clean-audit claim therefore holds precisely where there is no such sensor: in professions whose credentialing tests recognition and declarative knowledge but not production-class intervention under failure, and (the larger case, and this paper's central domain) in the fields with weak or absent licensure, no gated unaided examination, and no residency, where nothing between the entrant and the stock tests whether the entrant can produce when the tool is gone. §6 and §8.4 thus brace each other: medicine is the case where the credentialing layer catches the severance; the unregulated knowledge-work fields are the case where there is no such layer, and where a seat-binding instrument of the kind SI-WP-010 specifies (wherever such a present-engagement check is imposed, since these fields may have no licensing layer at all) reads clean while the bench empties.

The relationship to SI-WP-008 marks the extension rather than a reuse. SI-WP-008 describes ceremonial governance: oversight that *looks* substantive while being hollow. The failure this paper describes is its temporally displaced cousin and is not the same failure: the licensing here is *not* ceremonial, the human in the seat is genuinely engaged, the oversight is real. The problem is that the real substance is supplied by a stock the pathway may have stopped replenishing. SI-WP-008's failure is hollow oversight that was never substantive. This paper's failure is well-founded oversight resting on a foundation that is, on the conditional, being quietly depleted. A field cannot detect this by checking whether its oversight is ceremonial, because its oversight is not ceremonial; it can detect it only by measuring whether the substance currently present is being reproduced, which neither the licensing instrument nor the ceremonial-governance test looks at.

7. The Design Requirement: An Instrument That Measures the Flow

Following SI-WP-009's posture of specifying what any sufficient intervention must produce rather than prescribing a deployment, this paper specifies a design requirement and defers the architecture that would satisfy it. Providing the full implementation would convert a precondition-critique into a deployment roadmap, which is a different paper's job; the binding architecture for the flow is beyond this paper's scope and is left to later work.

Any binding architecture sufficient to preserve non-removability *over time*, as opposed to at a moment, must add a second instrument to the seat-binding instruments SI-WP-009 and SI-WP-010 specify. The existing instrument, licensing-as-architecture, asks: is substantive human engagement present in the work product now? It binds the seat and reads the stock. The missing instrument must ask: is the formative pathway that produces future substantive engagers intact? It must bind the flow.

The design requirement has one non-negotiable property, following directly from §3: the instrument must measure the **formation-trajectory of the bench, not the performance of the stock**, because the performance of the stock is the masked variable. An outcome audit (does the work product meet standard?) returns the health of the inventory and is structurally blind to the severance until stock-turnover. The instrument must therefore detect the *absence of the formative features* (§5.1) in the work assigned to entrants, because that absence is the leading indicator the outcome signal lags. Illustratively, and without prescribing implementation: whether entrants generate artifacts before consulting AI output or only check output already generated; whether entrants own downstream consequences or operate behind a senior's safety net that absorbs them; whether genuine self-generated-error struggle is present in the entry pathway or has been engineered out by AI scaffolding; what the ratio is, in an entrant's early-career work, of formation-bearing to formation-absent tasks; and, critically, whether entrants can perform the seat's intervention task *under conditions where the AI is absent or has failed*, not under conditions of "no AI ever," which would be anachronistic where AI is a permanent part of the workflow, but under the fallback conditions the seat exists to cover. That fallback-condition measure is the trajectory signal §3.1 identifies as the one operational metrics omit, and it is the same condition the §5.4 transfer study and the §8.6 credentialing question turn on. These measure *how juniors are being formed*, and their signal does not wait for the stock to turn over.

The instrument is also the empirical program. The same formation audit a governance architecture would use to bind the flow is the instrument by which this paper's central prediction could be *tested*, before the bench is reached rather than after. The design requirement and the falsification condition (§9) are the same artifact viewed from two directions, which is the strongest position a pre-empirical paper can occupy.

8. Boundaries, Counterarguments, and What Constrains the Claim

The paper engages the strongest rebuttals directly. Each constrains the claim; on the analysis here none defeats it; and the points at which a rebuttal *would* defeat it are stated.

8.1 The demand-shrinks rebuttal

If AI does more of the consequential work, perhaps the profession needs a *smaller* stock of human experts, and a thinner pipeline is correctly sized rather than severed. This is the most serious structural objection, and the response does not pretend to settle it. Severance versus correct-sizing is a quantitative question: a smaller stock still requires a positive flow, and a zero or near-zero flow fails *any* positive steady-state stock. A profession reaches a smaller stable size only if its reduced flow can still replace its reduced stock at the career-turnover rate. The

crisis bites only where flow-reduction outruns the reduction in needed stock. That ratio is the empirical pivot, it is currently unmeasured, and the structural argument does not establish its direction. What the paper claims is conditional and falsifiable: *where entry-level automation removes the formative periphery faster than automation retires the needed substantive stock, the steady state is unreachable.*

The strongest form of this rebuttal is architectural: an advanced system might keep only a small, central, elite tier of production-competent humans, route all hard cases to it, and staff everything else with verifiers, so the field needs the production-competent stock reproduced only at the center, not across the whole population. This sharpens the objection but does not dissolve the mechanism; it relocates it. Even a centralized elite tier must be *reproduced*, and if the flow that feeds it runs through verification-only roles that, on the hinge, do not form production competence, then the elite tier has no pipeline either: the severance is pushed up one level, not removed. The demand-shrinks rebuttal and the hinge of §5 are distinct routes to "no crisis," and they should not be conflated: demand-shrinks succeeds if the needed stock falls as fast as the flow (a ratio claim, independent of the hinge), and rerouting succeeds if verification forms the needed stock (the hinge). The paper's response to demand-shrinks is the ratio claim and stands on its own; its response to rerouting is the hinge; neither is presented as settling the other.

8.2 The market-self-correction rebuttal

Labor economics holds that firms *do* fund general training under realistic frictions (Acemoglu and Pischke, 1999), internalizing the value of the future stock. The competitive baseline runs the other way: Becker's (1962) result is that in a frictionless competitive market firms will not pay for general training, since the worker captures its return; it is precisely the labor-market imperfections Acemoglu and Pischke identify that overturn that baseline and make firm-sponsored general training an equilibrium outcome. If expertise reproduction were failing, firms would on that account have a private incentive to fund the formation that prevents it. The rebuttal is real and is defeated only by the masking claim, which makes masking load-bearing rather than incidental: firms do not fund formation against a deficit their instruments do not show. Output is rising, oversight is passing, total staffing surfaces look stable, and the masked variable is invisible to exactly the operational signals that would trigger the investment. The market does not self-correct against a failure its deployed measurements cannot see. If masking fails (if a firm's instruments do measure trajectory under fallback conditions) the incentive to invest is restored, which is why §8.3 matters and why the narrowed masking claim, not the broad one, is the one the paper defends.

8.3 The succession-planning rebuttal

The sharpest attack on masking is that ordinary workforce practice already observes pipeline gaps: succession planning, demographic forecasting, talent review, and "we have no one ready to promote" are routine, and they are explicitly designed to look past current performance at future potential. This is correct, and it bounds the claim. The response is not that these instruments are absent or badly run; it is an epistemic decoupling. Such instruments predict future capability through proxies: a junior who is excellent, fast, and reliable at the work in front of them scores as high-potential. Under severance, the work in front of the junior is verification of AI output, and the proxy, excellence at verification, stays internally consistent and highly correlated with the historical markers of potential while becoming *decoupled* from the target trait it is meant to predict, production-class competence under failure. The instrument is not malfunctioning by its own lights; it is measuring a trait that no longer maps to the competence the seat will require. A rational, well-run succession process can therefore read a healthy bench while the bench is forming into something the seat does not need. These instruments detect the

severance *only if* they measure demonstrated developmental trajectory under conditions where the AI is absent or failed, and where they do, masking does not hold and the field can see the problem, which the paper concedes directly, because conceding it is what keeps the claim honest. Where they instead read present performance on present (verification) work, they are decoupled from the variable that matters. The masking claim is therefore not that the gap is undetectable in principle; it is that the gap is invisible to the instruments most fields actually deploy, including their forward-looking ones, via decoupling, and visible only to an instrument of the kind §7 specifies, which measures fallback-condition trajectory and is not yet standard. A field with serious trajectory measurement under fallback conditions is, on this paper's own account, protected, and that is the recommendation, not a hole.

8.4 The tool-transition rebuttal, and the bounding case of regulated training

Calculators, compilers, spreadsheets, and CAD each automated work that was once formative, and expertise reproduction survived each; the historical base rate of "this tool will destroy a profession's ability to reproduce itself" is approximately zero, and the burden is on anyone claiming this time differs. The paper accepts the burden and meets it with a criterion, not a hedge, and the criterion must be made non-tautological, because "severing means it severs reproduction" explains nothing. Two distinctions do the work. The first is the *origin* of any replacement periphery. A tool is **organically relocating** if the new formative periphery emerges from the tool's own workflow at no special cost: the calculator removed arithmetic drill but left higher-order problem-setting; the compiler removed manual machine-code translation but left algorithm and system design. A tool is **structurally severing** if it destroys the old periphery and any replacement periphery must be *artificially reconstructed and subsidized* because the tool's economics do not provide one. The second distinction tests whether a candidate replacement periphery actually qualifies: a replacement periphery exists only if entrants can still perform tasks that are lower-risk than senior work, consequential enough to produce valid feedback, performed before exposure to the AI's answer, connected to eventual intervention competence, and abundant enough to form a cohort rather than a few exceptional trainees. A tool is reproduction-safe if it leaves or organically creates a periphery meeting that test; reproduction-severing if the remaining junior work fails it. The claim is not that AI is dangerous because it is new; it is that AI-mediated entry-level work, in the domains at issue, may meet the *severing* criterion, and the paper's burden is to show, domain by domain, that the tool eliminates rather than relocates a qualifying periphery. Where it relocates, the thesis does not apply.

Regulated professional training, with surgery and medicine the paradigm, is the paper's bounding contrast case rather than its central one, and the distinction matters because medicine is precisely the field where the mechanism is *most thoroughly countered*, not where it fails to apply. The mechanism applies fully: medicine is intervention-class and field-specific, so it satisfies the role criterion, and surgical residency satisfies the qualifying-periphery test of the paragraph above: supervised operating is lower-risk than unsupervised, consequential enough to produce valid feedback, performed before the trainee is handed the unaided case, connected directly to intervention competence, and run at cohort scale. Medicine is not a field where the formative periphery survived automation untouched; it is a field that *rebuilt* the periphery deliberately. It is also the field with the most developed countermeasures (simulation, residency, structured supervised progression, board certification, deliberately preserved training cases) and a referee who knows medical education would rightly say that medicine has spent a century building exactly the institutional substitutes the restoration-window inequality says are costly to stand up. That is the point of using it as the bound. Medicine did not preserve formation by default; it preserved formation by paying, deliberately and at sustained cost, for an institutionally subsidized formation periphery: residency and simulation are the artificially maintained formation periphery the origin distinction above contrasts with organic relocation. Surgical

training is therefore not a counterexample to the inequality; it is a worked instance of *beating* it, and the record of what beating it required is the evidence for what the inequality costs. The bounding case defines the safe edge: a field that recognizes the need before its stock turns over, and that has the institutional machinery and the will to pay, can preserve the flow. The fields this paper centers, unregulated professional knowledge work, have neither the machinery nor, absent a signal, the prompt to build it, and that is where the mechanism operates unchecked. Medicine shows what defeating the mechanism takes; its absence elsewhere shows why elsewhere is the worry.

8.5 What would make the paper wrong

The paper is a predicted failure mode conditioned on a contested antecedent, and the live possibilities under which it is mistaken are worth stating. If the four formative features turn out to be *present* in AI-mediated entry-level work, the severance premise fails and non-formation is not predicted. If verification-only experience *transfers* to production-class competence in intervention domains, the hinge fails. If firms' deployed instruments *do* detect the formation-trajectory deficit under fallback conditions before stock-turnover, masking fails and market self-correction is available. If severed pipelines prove *cheaply re-establishable* within the detection-to-exhaustion window, the inequality fails. If the relevant unaffected reservoir remains available at ecology scale, the severance is a hiring problem rather than reproduction failure. If the displacement antecedent does not hold, if entry-level work is not in fact being severed, the mechanism has no occasion to operate. None of these is currently ruled out, and the antecedent in particular is genuinely contested (§9).

8.6 The credentialing gatekeeper

The credentialing rejoinder of §6 deserves its own treatment, because it is both the strongest detection mechanism the masking argument must survive and, handled correctly, a further arm of the thesis. Many regulated professions require new entrants to demonstrate competence before admission to the stock. Where that demonstration tests production-class intervention under fallback conditions (open-ended generation under uncertainty, not declarative recall or clean standardized scenarios) the credentialing examination is a partial version of the §7 flow-instrument, and a severed flow would show up as falling readiness or pass rates before the stock turned over. So the masking claim is bounded: it is weakest exactly where credentialing robustly tests fallback-condition production, and strongest where credentialing tests recognition and knowledge but not production under failure, or where there is no credentialing gate at all.

Two observations extend rather than retreat the claim. First, most credentialing, even in gated professions, tests declarative knowledge, protocol adherence, or supervised standardized cases, not the "AI is absent and the system is failing" condition the seat exists to cover; a gate that does not test the fallback condition does not catch a severance in fallback competence. Second, and more pointedly, the credentialing signal is itself subject to the same dynamic: as AI tools become embedded in education, preparation, and examination, candidate performance can be inflated by scaffolding that will not be present at the moment of genuine fallback, so the gate that was supposed to be the independent check on the flow is co-opted by the very mechanism it would detect. Where the gate robustly tests unaided production, the field is protected and the credentialing-sensor story of §6 holds, and this is medicine's early warning; where the gate is weak, absent, or AI-co-opted, the condition of most professional knowledge work, the masking logic stands and the gate provides no rescue.

9. Epistemic Status and Falsification Conditions

9.1 Epistemic status

The mechanism (§3), the observability extension (§4), and the blind-spot argument (§6) are asserted at strength, because they are structural: they follow from the separability of stock and flow and from the triple identity of the self-consuming mask, and hold as relations *if* the antecedent obtains. A word on the restoration-window inequality specifically, because the strength claim and §3.3's own hedging must be reconciled: what is structural is the inequality's *terms, direction, and measurement target*: given the antecedent, restoration cost and remaining formation capacity are the variables whose relation determines whether a window closes, and that relation follows from the three terms. Whether, and where, that window falls in a given field is the quantitative question §3.3 leaves open, depending on turnover, production-function shape, periphery elimination, preserved teaching capacity, and reservoir availability. The structural claim is the form of the inequality and what it makes the load-bearing variables; the empirical question is whether, and where, the crossing point falls. These are different postures and the paper holds both, not one in place of the other.

What is calibrated to its evidence, and is not asserted at structural strength, is the antecedent and the forward outcome.

The **displacement antecedent**, that entry-level work is being automated and the formative pathway thereby severed, is inherited from SI-WP-009 and SI-WP-010 and is empirically contested. The evidence is early and the magnitude disputed. The Stanford "Canaries" analysis, drawn from ADP payroll data, reports a relative employment decline for early-career workers in AI-exposed occupations; Anthropic's 2026 analysis, drawn from Current Population Survey data, finds suggestive evidence that hiring of younger workers into exposed occupations has slowed, a result it flags as barely statistically significant and open to alternative interpretation. The two sources disagree on magnitude and confidence; where they agree, it is on *locus*, that the visible margin, if any, is entry and hiring rather than incumbent displacement, not on whether the effect is real. A third source converges on the same locus by a different method: the 2026 Oliver Wyman Forum and New York Stock Exchange survey of 415 chief executives reports that the share planning to deprioritize hiring for junior roles over the next one to two years rose from 17% to 43% year over year, accompanied by a shift toward midlevel roles and flat total headcount, the talent pyramid reshaping toward a middle-heavy form, which is the headcount split of §3.1 appearing as a planned structural change rather than an inference. Two caveats hold this in the contested-antecedent register and must not be dropped. First, it is stated intent over a one-to-two-year horizon, not realized severance, and the survey frames every forward figure as a plan rather than a measured outcome; intent is evidence about the direction and locus of pressure, not about whether the formative pathway has in fact been cut. Second, the same survey reports a contrarian subset, the most advanced AI adopters shifting *toward* junior hiring at a higher rate than laggards, on the view that AI raises rather than lowers the value of entry-level talent, which cuts against any uniform reading of severance and is recorded here for that reason. Taken together, the payroll, survey, and CEO-intent sources triangulate by three independent methods on the same conclusion: the margin under pressure, to the degree there is one, is the entry and formation layer rather than the incumbent stock, and by CEO account that pressure is intensifying. None of the three establishes that the pathway has been severed. Against them sit cross-country nulls and remote-work confounds. This paper does not adjudicate that literature, does not relitigate it, and does not assert that severance is presently established; it specifies the mechanism that obtains *if* the antecedent holds and inherits the antecedent's contested status openly.

The **non-formation claim**, that entrants will fail to acquire production-class competence where AI performs the formative work, is the paper's central forward claim, and its status must be stated exactly, because it is easy to overstate. At minimum it is a structurally predicted risk, following from §5 given the antecedent; that floor does not depend on the empirical literature. To the extent the professional-education literature has independently named and discussed it, under the label "never-skilling" (Ke et al., 2026; Oettl et al., 2026) and the convergent label "upskilling inhibition" (Natali et al., 2025), distinguished in each case from ordinary expert deskilling and flagged as a risk warranting longitudinal study, it is also independently recognized, which raises it above an author-coined concern. That literature establishes *conceptual uptake*, the field judging the mechanism coherent enough to name and warn about, not *demonstrated occurrence*. The empirical base is sparse, indirect, and mixed: reviews identify plausible mechanisms and find that evidence for AI fostering higher-order reasoning is less robust than for efficiency and basic-knowledge gains, while at least one small study of AI-supported radiology-resident training found performance *improved* and trainees resilient to AI error (Insights into Imaging, 2025). The honest ceiling is therefore: non-formation is an independently recognized, contested risk construct with early and mixed support, not an established causal effect. The completed CVP/SF0037 pass confirms this status for publication: the distinctness of non-formation from deskilling was verified across multiple sources, so the claim holds in the named-but-contested register rather than falling to the structural floor. Should later literature prove thin or collapse that distinction, the claim downgrades to the structural floor; the distinctness from deskilling is load-bearing, because SI-WP-008 owns deskilling and this paper's novelty and its calibration both depend on the separation. The claim's strength rests on the structural argument and on the falsifiable predictions below, held under the corpus's pre-empirical posture. The paper does not move from "the risk is named" to "the phenomenon is observed"; that step is not licensed by the evidence and is not taken anywhere in §§3–7, whose arguments are conditional by construction and do not depend on it.

9.2 Falsification conditions

The paper can fail in two distinct ways, and separating them keeps the epistemic posture honest. The conditional **mechanism** can be refuted: shown not to follow or not to operate even where the antecedent holds. Or the **antecedent** can fail to instantiate, in which case the mechanism remains structurally valid but has no occasion to operate in the domain at issue. The first is refutation; the second is non-instantiation. They are not the same, and a conditional paper is not "wrong" merely because its antecedent is absent in some field.

Mechanism-refuting conditions:

1. **The transfer study.** If, in production-class-intervention domains, verification-only experience produces intervention competence equal to production-formed experience at genuine system-failure points, the hinge (§5) is refuted and the flow is rerouted rather than severed.
2. **The formation-feature audit.** If AI-mediated entry-level work *preserves* the four formative features, most decisively if industry-standard entry roles come to mandate an unaided blind-generation phase before AI exposure as a non-negotiable part of the work, the severance premise fails, because the formative rep has been retained inside the verification era. (Note this is a workflow-evolution condition, not the tautology that "if the work is not schema-bypassing then the premise fails": it specifies an observable institutional change that would preserve formation.)
3. **The detection counterexample.** If the instruments fields actually deploy reliably detect formation-trajectory deficits under fallback conditions before stock-turnover, the masking claim (§3.1, §8.3) fails.

4. **The reversibility counterexample.** If severed formative pipelines prove re-establishable within the detection-to-exhaustion window at modest cost, the restoration-window inequality (§3.3) fails.

Instantiation-defeating conditions:

5. **The ratio.** If flow-reduction tracks or lags the reduction in needed substantive stock, the severance reframes as correct resizing (§8.1) and the mechanism, though valid, does not bind.

6. **The reservoir.** If an unaffected reservoir remains available at ecology scale, the severance is a hiring problem rather than reproduction failure (§3.3).

7. **The antecedent.** If entry-level developmental work is not in fact being severed, the mechanism does not operate, whatever its structural validity.

The structural core (§3, §4, §6) is analytic given the antecedent; it is "wrong" only via a validity challenge (that the relations do not in fact follow, for instance that self-masking does not follow from stock/flow separability and the triple identity) rather than by empirical falsification, and the paper invites that challenge on the derivations directly. The conditions above falsify the empirical layer; the validity of the core is contestable on the reasoning, and the paper's "conditions under which it is wrong" cover both.

9.3 Research agenda

The instrument specified in §7 is the empirical program. Building formation-feature audits and fallback-condition trajectory measures is how a field could detect a severance before the bench is reached, and the same audits are how mechanism-refuting conditions 1, 2, and 3 would be checked. The corpus's contribution at this stage is not a finding; it is the specification of *what to measure, and why the obvious thing to measure, present output, is the wrong thing*, together with falsifiable predictions that would convert the structural argument into an empirical result.

10. Conclusion

SI-WP-009 preserves the seat. This paper asks who will be capable of filling it, and names the mechanism by which the answer could become *too few*, too few qualified occupants in the domains where the conditions bind, without the instruments a field ordinarily runs registering it in time: a severed flow masked by an intact stock, where the mask is the very inventory being drawn down and is also the irreplaceable component of the means to rebuild, and where the formative work novices needed has been removed as human work rather than reassigned, so that the pathway grows costlier to rebuild as the window to rebuild it closes. On that mechanism, and conditioned on an antecedent the paper does not claim to have established, a field could hold a substantively capable human in every consequential seat, pass every present-engagement audit it runs, post rising productivity, and be failing to reproduce its expertise at the same time, with the corpus's own seat-binding architecture reading clean throughout, because that architecture measures the seat, and the seat is the variable the failure leaves intact. The failure is not invisible in principle: it is invisible to the instruments that measure present performance, and visible to one that measures formation trajectory under fallback conditions, which a field running such measurement can see coming, and most do not run. The non-removability prescription is necessary and temporally insufficient. The instrument that would close the gap measures the formation of the bench, not the performance of the stock, and the time to build it is before the field reaches for the bench, not after it finds the bench thin.

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- Note on terminology: Ke et al. (2026) and Oettl et al. (2026) name the novice-side risk "never-skilling"; Natali et al. (2025) draw the same novice-side distinction under the label "upskilling inhibition." The construct is convergent across venues under both terms; this paper uses "never-skilling," noting the sibling term.*

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